

Oregon — ensemble calibration targets

Project: or-spatial-targets · 16,814 HRUs · PNW · rebuilt 2026-09-09

The target NetCDFs now carry the **per-source ensemble** alongside the bounds, and recharge / soil moisture normalize **each source over its own period of record**. This deck covers what changed, what the new files contain, and how to read them.

USGS National Hydrologic Model · TM 6-B10 (Hay et al. 2022) · issue #338 · supersedes the May aggregated + target overview for this fabric

What changed

	Before	After
Variables per target NC	3 — lower_bound , upper_bound , n_sources	8 — the three above + one variable per source + ensemble_mean + ensemble_std
Provenance attrs	source (prose)	+ member_keys (machine-readable, emitted set) · + members_emitted
Conventions	CF-1.6	CF-1.8 on target NCs (consolidated + aggregated stay CF-1.6)
Recharge / soil-moisture normalization	one shared window for every source	per_source_por — each source over its own complete-calendar-year record
SWE sources (OR)	5 (incl. Daymet)	4 — Daymet dropped
SCA (OR)	built, in the target set	still built ¹ , excluded from the ensemble narrative

`emit_members` is an output switch, not a science switch. Members were always computed — they are the input the bounds are derived from. The flag only decides whether they are also written. `lower_bound` / `upper_bound` are byte-identical either way.

¹ `enabled: false` is honored when `nhf-targets run` selects all targets, but **not** when a target is named directly — and the SLURM array names them (`run-sca`). So `sca_targets.nc` was rebuilt on 2026-09-09 despite being disabled in `config.yml` . It carries no members (`members_emitted=false`), correctly: SCA's bounds are a MOD10C1 confidence interval, not a member min/max, so emitted members would not reconstruct them.

Oregon at a glance — rebuilt 2026-09-09

Target	Members (from the NC's member_keys)	Period	Steps	n_sources max
Runoff	era5_land · gldas_noah_v21_monthly · mwbm_climgrid	1979-01 .. 2024-12	552	3
AET	mod16a2_v061 · ssebop · mwbm_climgrid	2000-01 .. 2024-12	300	3
Recharge	reitz2017 · era5_land	2000 .. 2013	14	2
Soil moisture	merra2 · nldas_mosaic · nldas_noah	1980-01 .. 2024-12	540 + 45	3
SWE	snodas · era5_land · margulis_wus_sr · ua_swe	1980-01 .. 2024-12	16,437	4
SCA	(none — CI-interval bound)	1982-01 .. 2024-12	15,706	2

File	Was	Now
runoff_targets.nc	66 MB	212 MB
aet_targets.nc	33 MB	111 MB
soil_moisture_targets_monthly.nc	41 MB	155 MB
recharge_targets.nc	2.0 MB	5.1 MB
swe_targets.nc	736 MB	3.1 GB

Every member list above is read back from the written file's member_keys attr and cross-checked against the variables actually on disk — not from config.yaml . Growth is the members plus the two ensemble statistics. The _nn_filled companions did **not** grow (33 / 66 / 41 / 2.1 MB): by design they carry only the filled bounds, since filling an individual member would fabricate a source observation where that source never had one.

Per-source period-of-record normalization

Recharge and soil moisture are normalized to 0–1. Previously every source shared one window — which clamped a long record to the shortest one. Under `normalize_period: per_source_por` each source uses its own complete-calendar-year record:

Target	Source	Window actually used
Recharge	reitz2017	2000-01-01 / 2013-12-31
	era5_land	1979-01-01 / 2025-12-31
Soil moisture	merra2	1980-01-01 / 2025-12-31
	nldas_mosaic	1979-01-01 / 2025-12-31
	nldas_noah	1979-01-01 / 2025-12-31

Under the old shared window both recharge sources were clamped to 2000–2013; ERA5-Land now normalizes against its full 47-year record. MERRA-2's one-year-later start is detected automatically.

"Period of record" means complete calendar years. A partial leading or trailing year is coverage, not record — an unfinished year's annual *sum* would otherwise become that HRU's minimum and compress every other year toward 1.0.

Each window is recorded on the output NC as `normalize_window_<source_key>`, so the file states the record it actually used. Two traps had to be avoided: reindexing pads uncovered months with NaN at *real* timestamps (so a naive completeness count reads padding as data), and resampling to annual collapses a ragged partial year into one indistinguishable step. Both windows above are derived from the raw, pre-reindex, pre-resample series. No interior-gap warning

What the ensemble buys you

Each target NC now carries every contributing source as its own variable, plus two derived diagnostics:

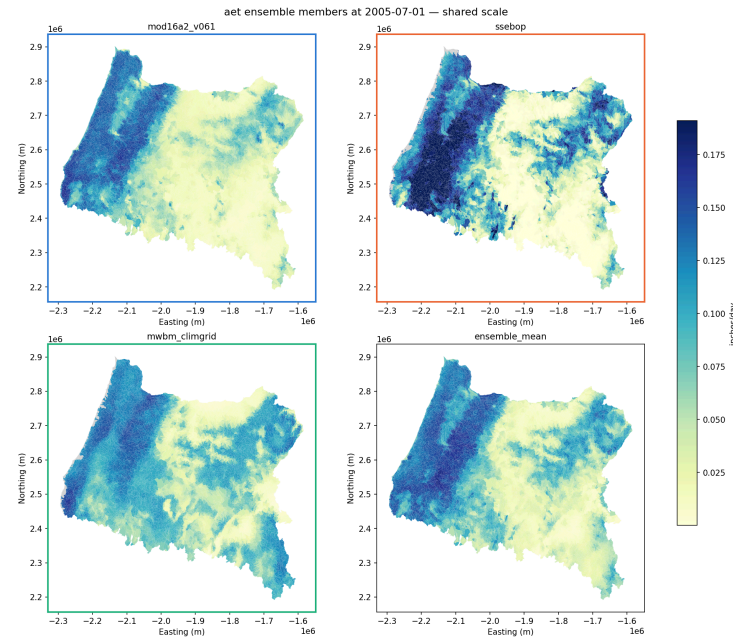
Variable	Meaning
<source_key>	that source's harmonized contribution, in the target's units
ensemble_mean	NaN-aware mean across finite members
ensemble_std	population std across finite members — NaN wherever <code>n_sources < 2</code>
lower_bound / upper_bound	unchanged: NaN-aware min / max across members
n_sources	unchanged: count of finite members per (HRU, time)

Previously, a wide bound told you the sources disagreed but not **which** source drove the edge. The members answer that directly, per HRU per timestep.

`ensemble_std` is masked below two sources deliberately. With one finite source the population std is exactly 0 — a calibration weight built from it would read "only one source covered this cell" as "perfect inter-source agreement."

Median `ensemble_std` where defined: runoff 0.43 cfs (99.1% of cells) · AET 0.0119 in/day (95.1%) · recharge 0.085 (99.6%) · soil moisture 0.109 (98.0%) · SWE 0.00068 in (95.0%). SWE's median is near zero because most (HRU, day) cells are snow-free and every source agrees on zero — read it seasonally, not as a single number.

Seeing the members



AET members and `ensemble_mean`, July 2005 — one shared color scale

The shared scale is the figure. Autoscaling each panel to its own range

would render three sources that disagree by a factor of three as three

near-identical maps. The scale is pooled across every panel

instead, so a

systematically wet or dry source is visible at a glance

Coverage gaps stay grey rather than reading as zero.

`mod16a2_v061` has no

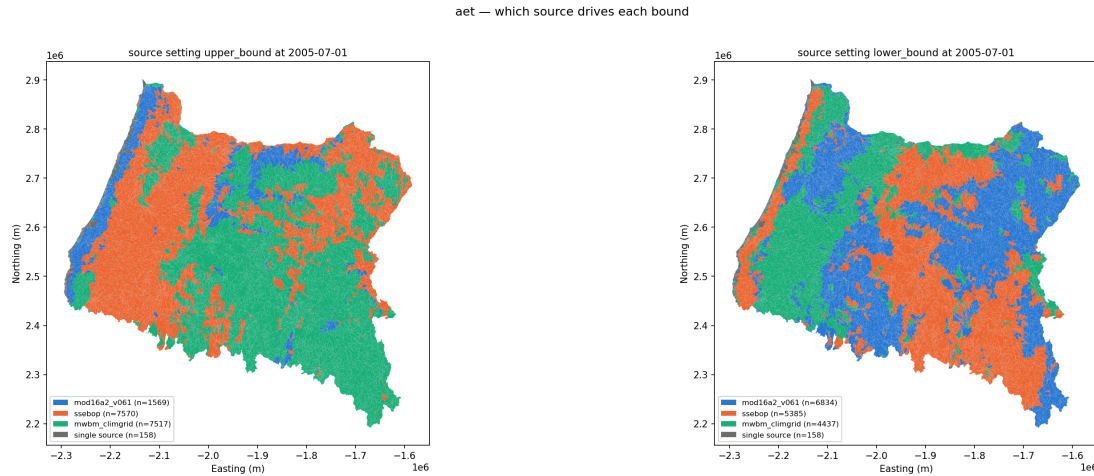
data on the coastal fringe — the same 158 HRUs that sit at

`n_sources == 1`

there, where the bound is degenerate.

Panel frames carry each member's color, reused in the two

Which source drives each bound



AET, July 2005 — upper_bound (left), lower_bound (right)

upper_bound is the NaN-aware max across members, so exactly one member sets it at each HRU. This names it.

- mod16a2_v061 sets the upper bound at **1,569** HRUs but the lower at **6,834** — systematically the driest July AET source.
- ssebop (7,570) and nwbm_climgrid (7,517) split the upper bound almost evenly, trading along the Cascade crest.
- Recharge is starker: reitz2017 sets the upper bound at **14,908** of **16,814** HRUs and era5_land the lower at the same count.

Grey is not a member. **no spread** = the sources agree; **single source** = only one is finite, so there is nothing to win.

A driver map is a trap without a tie test

argmax always returns a column — even when every member holds the same value. Members reach the figure as float32 that has been through a unit conversion, so physically identical values are rarely bit-identical.

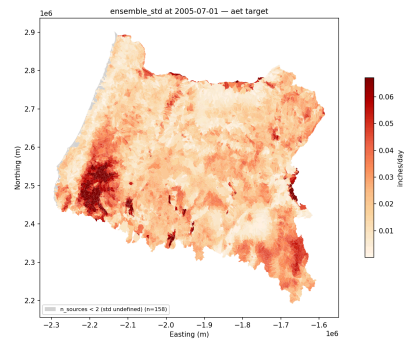
Oregon SWE, 16,814 multi-source cells, **2010-08-15** (snow-free):

	cells
spread bit-identical zero	239
0 < spread < 1e-9 inches	14,239
field maximum that day	108 inches

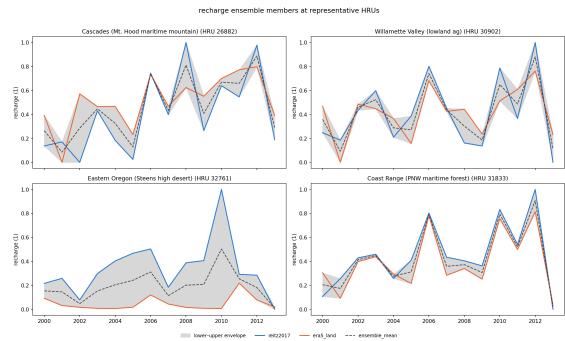
An exact spread == 0 test hands **85% of a snow-free state** to a driver chosen by a nanometre of SWE. The tie test is therefore spread <= atol + rtol · max|value| , scaled by the field rather than the cell — near-zero cells have no magnitude of their own to be relative to, and that is exactly where the noise lives.

upper_bound	classified "no spread"	exact	scale-aware
2010-08-15 — snow-free		1.4%	89.0%
2010-03-01 — near-peak snow		1.3%	17.2%

Where, and when, they disagree



AET ensemble_std , July 2005



Recharge members at four representative HRUs

AET disagreement concentrates in the Coast Range and the west slope of the Cascades — forested, orographic, and where ET products diverge most. The legend names the grey: 158 HRUs at `n_sources < 2` , where a population standard deviation would be exactly `0` and read as perfect agreement.

In time, the same question resolves per HRU. At Steens the two recharge sources hold a sustained offset across the whole record; in the Coast Range they track each other closely.

A bound that hugs one colored line through a season is a bound set by that source. That is the read the bounds alone could never give you.

Coverage — where the bound is actually informative

n_sources distribution across all (HRU, timestep) cells:

Target	0	1	2	3	4
Runoff	—	0.9%	54.3%	44.8%	—
AET	—	4.9%	12.5%	82.6%	—
Recharge	0.0%	0.4%	99.6%	—	—
Soil moisture	—	2.0%	—	98.0%	—
SWE	0.1%	5.0%	12.3%	46.2%	36.5%

- **Runoff** sits at 2 sources more often than 3 — MWBM ClimGrid ends 2020, and GLDAS starts 2000, so the full three overlap only 2000–2020.
- **AET** is the best-covered monthly target at 82.6% three-source.
- **SWE** reaches 4 sources on 36.5% of cells; Margulis contributes by geometry (honest NaN outside its Western-US grid) and SNODAS starts 2004.
- A cell at n_sources == 1 has a **degenerate bound** — lower equals upper. Treat it as a point estimate, not a range.

Percentages are of all (HRU, timestep) cells including snow-free and out-of-coverage ones, computed directly from each rebuilt NC's n_sources variable.
Recharge's 0.0% at zero sources is a genuine non-zero count that rounds to zero at one decimal.

SWE — Daymet dropped, four sources

`snodas` · `era5_land` · `margulis_wus_sr` · `ua_swe` — Daymet removed per the colleagues' ensemble notebook; UA SWE retained at their request.

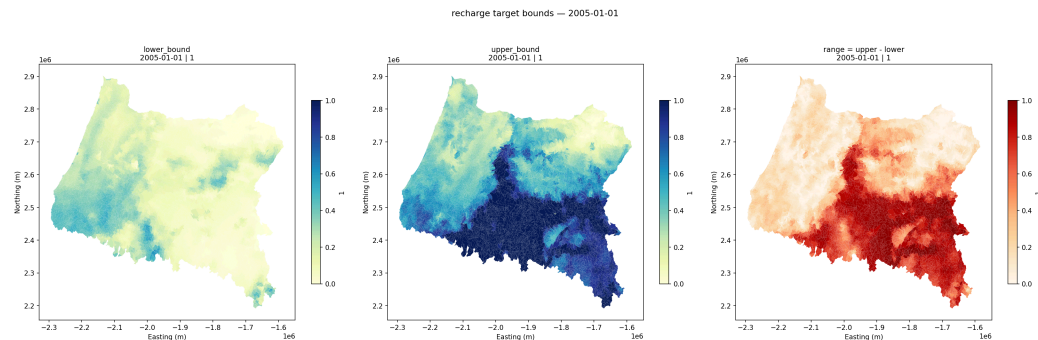
Source	Contributes	Note
<code>snodas</code>	2004-	21 per-year NCs; the archive has real day-gaps in 2004-07
<code>era5_land</code>	1979-	47 per-year NCs; <code>sd</code> is snow depth water equivalent , so $\times 1000 \rightarrow$ mm is exact, not a density assumption
<code>margulis_wus_sr</code>	1985-2020	36 per-year NCs; 500 m, Western-US only — contributes by geometry, honest NaN elsewhere
<code>ua_swe</code>	1982-2022	41 per-year NCs; CONUS-wide, reaches before SNODAS

Bound range across the rebuilt file: **0 - 288.9 inches**.

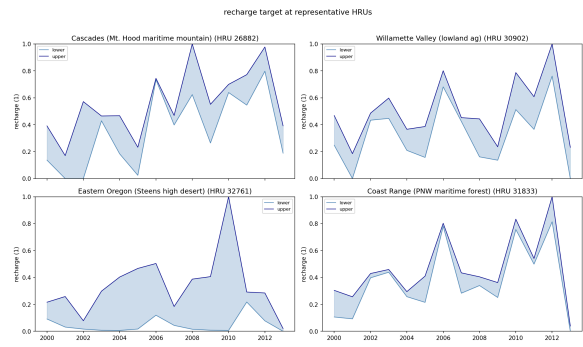
A year in which a source has no data still emits that source as an **all-NaN member**, so every per-year file carries the same variable set — the stitch across 45 years requires an exact match. Getting this subtly wrong is what broke the first rebuild attempt (#345).

The written `lower_bound` minimum is `-2.9e-23` inches — floating-point noise from the unit conversion, numerically zero, not negative snow. Worth knowing before someone reports it as a bug. Member counts above are the per-year NC counts the build logged, which match each source's expected coverage exactly.

Recharge — per-source POR, visible



Bound width per HRU, 2000–2013



Representative HRUs — normalized 0–1

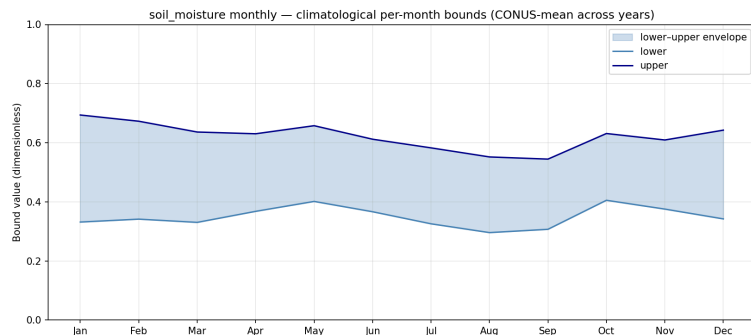
Both sources are normalized to 0–1, but now over **different windows**:

- reitz2017 → 2000-01-01 / 2013-12-31
- era5_land → **1979-01-01 / 2025-12-31**

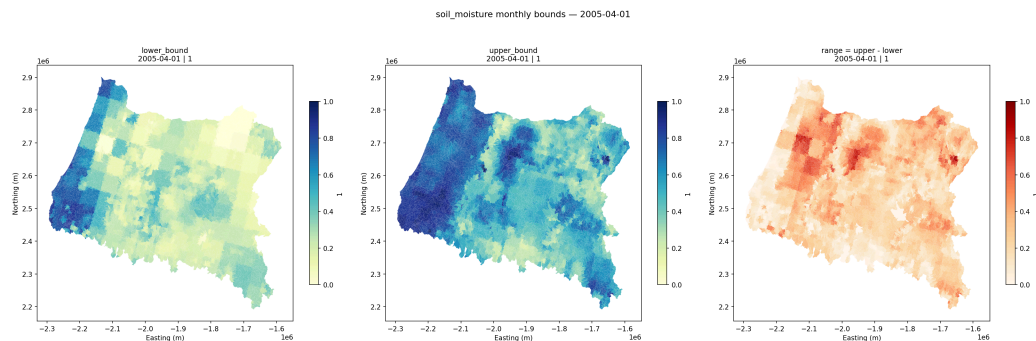
Previously both were clamped to 2000–2013, so ERA5-Land's 0 and 1 were set by the wettest and driest year *in the Reitz window* rather than in its own 47-year record.

These figures changed against their June versions. Runoff and AET re-rendered byte-identical — neither their sources nor their method moved.

Soil moisture — per-calendar-month, per-source window



Monthly climatology of the bound



Bound width per HRU

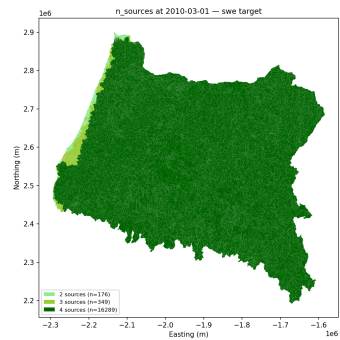
Normalization is **per calendar month** — a January value is scaled against other Januaries, so the seasonal cycle does not swamp the interannual signal.

Under `per_source_por` each source brings its own window:

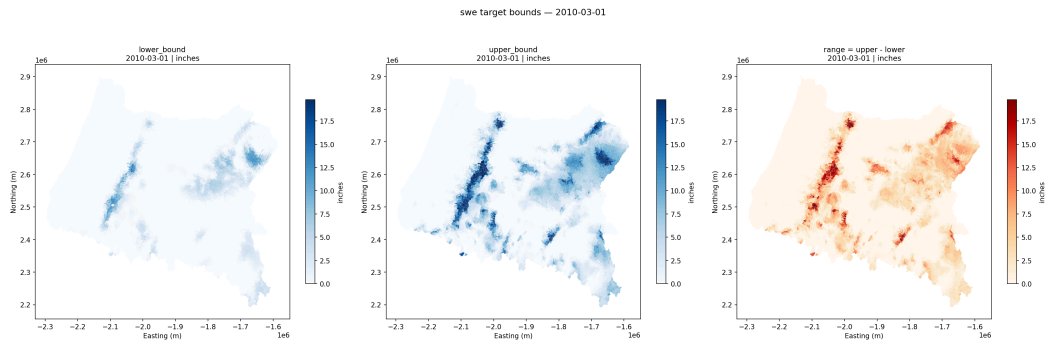
- `merra2` → **1980**-01-01 / 2025-12-31
- `nldas_mosaic` → 1979-01-01 / 2025-12-31
- `nldas_noah` → 1979-01-01 / 2025-12-31

MERRA-2's one-year-later start is detected from its own record rather than being imposed on the other two.

SWE — four sources, not five



n_sources per HRU — max 4



Bound width per HRU, inches

Daymet dropped per the colleagues' notebook;
ua_swe retained at their request.
snodas · era5_land · margulis_wus_sr · ua_swe .

The n_sources map is the direct evidence — its maximum is now **4**. Coverage across all (HRU, day) cells: 4 sources at **36.5%**, 3 at **46.2%**, 2 at 12.3%, 1 at 5.0%.

Margulis contributes by **geometry** — area-weighted inside its Western-US grid, honest NaN outside — which is why the 4-source region is spatially bounded rather than uniform.

A cell at n_sources == 1 is a **degenerate bound**: lower equals upper. Read it as a point estimate.

Reading the new files

```
import xarray as xr
ds = xr.open_dataset("targets/swe_targets.nc")

ds.attrs["member_keys"]      # 'snodas,era5_land,margulis_wus_sr,ua_swe'
ds.attrs["members_emitted"]  # 'true'
ds["snodas"]                 # that source's contribution, in inches
ds["ensemble_std"]           # NaN where n_sources < 2
```

- **member_keys is the emitted set.** `source_keys` is a *different* attr with different semantics — the sources the config requested — and the release publish gate compares it to `config.yml` with strict equality. Do not conflate them.
- **members_emitted is always present**, so a reader can distinguish "this target never emitted members" from "members were emitted and something dropped them."
- The `_nn_filled` companion carries **only** the filled bounds and stamps `members_emitted=false`.

Turning `emit_members: false` on a target changes no bound value — only file size. On the national fabric a daily SWE target goes from ~12 GB to ~36-48 GB with members, which is why the switch exists.

Caveats and what is not done

- **Figures here are current** — re-rendered 2026-09-10 against the rebuilt targets, adding the four ensemble figures. Every *map* also changed cosmetically: the axes were labelled "Longitude"/"Latitude" while the Oregon fabric is EPSG:5070 Albers, so every OR figure had been reporting metres as degrees. They now read `Easting (m)` / `Northing (m)`. The May OR deck carried superseded plots and is removed on this branch.
- **The member and spread figures now exist** (issue #351): member small multiples, an `ensemble_std` map, driver maps, and per-member series, in all five member-carrying notebooks. Building them surfaced a real bug — the first tie test compared spread against exactly `0`, which left 85% of a snow-free August attributed to a noise-chosen driver; see the tie-test slide above. `snow_covered_area` gets a note instead of figures: its bounds are a MOD10C1 confidence interval, not a member min/max.
- `sca_targets.nc` **was rebuilt despite** `enabled: false` and is in the manifest as a published target (accepted for now).
- **Aggregated-figure notebooks glob disk, not config.** They will still render a Daymet panel for SWE even though Daymet is no longer a member of the target. Read the source of truth from the target NC.

Open follow-ups: #339 release payload stages build intermediates · #340 per-year target layout · #341 notebook feedback for the source notebook · #342 CF `time: point` alongside `time_bnds` · #343 versioning project intent artifacts · #347 test fixtures diverge from what the aggregator writes.

References

- **Issue #338** — ensemble members + per-source POR normalization · **#345** — the all-NaN member coord regression
- **Source notebook** — colleagues' 06_Create_calibration_target_ensembles.ipynb
- **TM 6-B10** (Hay et al. 2022) — docs/references/tm6b10.pdf ; this pipeline differs deliberately on time windows and dataset versions
- **Design + plan** — docs/superpowers/specs/2026-09-08-ensemble-target-schema-design.md , docs/superpowers/plans/2026-09-08-ensemble-target-schema.md
- **Architecture** — docs/architecture/transformation-pipeline.md (per-source POR, member emission), docs/architecture/nc-encoding-policy.md

Every number in this deck was read back from the rebuilt target NetCDFs on 2026-09-09, not from configuration.